AetherGrid

Cold-chain intervention software with explicit simulation boundaries

Author and scope

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- Professional role is author context only; no employer sponsorship or endorsement is claimed.

What the project demonstrates

- Detect heuristic distress and qualify a compatible buyer.
- Reserve simulated resources; require operator confirmation.
- Record an auditable simulated ledger, not a physical rescue.

Praveen Sudireddy | Digital Specialist Engineer (DSE), Infosys

An alert is not a completed intervention.

A qualitative engineering problem, not a market-size or national-loss claim.

Questions to resolve

- Who can authorize a cargo redirect?
- Is a buyer compatible, reachable and available for the full lot?
- How are reservations, confirmation and receipt reconciled?

Project approach

- Connect telemetry, model state and reviewed transaction workflow.
- Reject unsuitable bids and release expired reservations once.
- Keep failed lots and uncertainty visible; never treat an award as delivery.

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A narrow, supervised mango use case.

Bengaluru–Hyderabad is a proposed pilot corridor, not an office or deployment claim.

Proposed pilot boundary

- Mangoes; 5–10 supervised trucks only after access and consent.
- Manual operator decisions with a no-money ledger.
- Existing telemetry/log exports may be used; hardware fabrication is excluded.

Not established

- No observed field pilot, validated quality forecast or measured benefit.
- No real bank escrow, UPI or insurer execution.
- Legacy NH-44 map, buyers and carriers are demonstration data.

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Five layers, with a visible provider boundary.

React portals → native ASP.NET workflow → SQLite documents and durable outbox.

L0–L2

• L0: software telemetry provider and local simulators.

• L1: software HMAC v2, identity binding, time and sequence checks.

• L2: BioTimer, thermal Dijkstra, buyer eligibility and settlement arithmetic.

L3–L4

• L3: transactional reservations, reviewed settlement and reported receipt.

• L4: Command, Buyer, Analytics, Carrier and Insurance portals.

• Python remains a reference/demo implementation; native .NET uses HTTP + SSE.

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A buyer reserves. An operator confirms.

Browser-tested with separate authenticated buyer and operator accounts.

Initiation

- Record an operator-reviewed initial biological baseline.
- Accept signed telemetry and open a time-boxed distress auction.
- Check full lot, policy, capacity, cash, radius, reachability and deadline.

Completion controls

- Reserve simulated funds/capacity atomically, without settlement.
- Operator confirms/rejects; timeouts release once with bounded fallback.
- Reported full-lot receipt releases capacity once; disputes require review.

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Q10 is a heuristic, not a food-safety forecast.

No trained ML model or independently validated prediction accuracy is claimed.

BioTimer calculation

• Rate = Q10^((T−5)/10) × humidity factor × ethylene factor.

• Consumed life += rate × exposure hours / reference life; door-open factor is 5.

• Mango parameters: 264 h reference life, Q10 2.8, optimal 12°C.

Interpretation and limits

• Optimistic RVH differs from current-temperature spoilage ETA.

• Native receiver persists checkpoints and uses signed elapsed time.

• Parameters, maturity, handling and quality endpoints need real-data review.

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The stated baseline gives 90.65% recovery.

Synthetic 1,000 kg mango lot; ₹120/kg contract baseline; ₹78/kg rescue bid.

Explicit arithmetic

- Contract value ₹1,20,000; simulated salvage ₹78,000.
- Sum assured ₹72,000; cover cap ₹32,400; deductible 5%.
- Calculated insurance credit ₹30,780; buyer discount 35%.

Outcome classification

- Farmer simulated ledger credit: ₹1,08,780.
- ₹1,08,780 / ₹1,20,000 = 90.65%; not a universal guarantee.
- No real payment, insurer acceptance or independent physical receipt.

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Rank delivery, liquidation and cold holding.

Reference-algorithm parity is distinct from validated navigation.

Implemented native planning

• Bidirectional thermal Dijkstra over a fixed demonstration graph.

• Costs include temperature, reefer state, coverage, road/risk and life urgency.

• Operational buyer filters add policy, commodity, capacity and model reachability.

Decision boundaries

• 70 cases / seven profiles: 2,100 fields matched the Python ranked plans.

• Optional automatic model target selection never dispatches a physical truck.

• Cold holding may slow decay; it never resets consumed biological life.

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Executed checks, with their actual scope.

Fresh builds and isolated synthetic databases; results are not field efficacy.

Regression and native services

• 51 Python regression tests passed.

• 2,230 existing BioTimer/settlement/eligibility parity assertions passed.

• Published native build, signed telemetry and transactional HTTP checks passed.

Real browser journey

• Buyer login → reserve; separate operator login → confirm.

• UI shows ₹1,08,780; another buyer cannot read that award or totals.

• Five portals load; receipt/duplicate release and CSV reporting checked through API.

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Every rupee needs a stated baseline.

Deterministic simulation arithmetic is not insurance adjudication.

Rules

• Insurance = min(shortfall, cover × assured amount) × (1−deductible).

• Farmer total includes salvage, calculated insurance and bounded penalty top-up.

• A target floor is not guaranteed when the available top-up is insufficient.

Evidence and reconciliation

• Confirmed settlements retain an evidence snapshot and hash.

• Provider sandbox obligations use integer paise and explicit idempotency keys.

• Terminal callback changes, wrong amounts and stale events are rejected.

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Software integrity is not sensor truth.

No hardware signing, device mTLS or physical-tamper proof is implied.

Implemented controls

• Opaque HttpOnly sessions; PBKDF2 hashes; role and buyer/device bindings.

• CSRF, same-origin mutation checks, login throttling and request limits.

• Persisted replay cursor, password recovery, session revocation and key rotation.

Boundary and remediation

• Acceptance review found and fixed a buyer-role portal access defect.

• Buyer summary, auctions and settlements are now scoped to that identity.

• Host compromise, external pentest, MFA/SSO and TLS deployment remain separate risks.

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Crash recovery must preserve obligations.

Forced-process termination was tested with a pending synthetic reservation.

Durable behavior

• SQLite WAL transactions persist state with a sequenced event outbox.

• SIGKILL/restart: pending resources released once after deadline.

• Stale telemetry alerts and expired-authentication cleanup were exercised.

Safe operations

• Retain telemetry, ledger, evidence, callbacks and replay cursors.

• Use online SQLite backup; rehearse restore before reopening ingestion.

• A restored old backup may lose recent replay watermarks; reconcile before use.

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Learning, rehearsal and auditable decision support.

Personal engineering work; no paid traction, commercial forecast or partnership claim.

Useful applications

• Explore how telemetry can initiate a reviewed cold-chain workflow.

• Study full-lot eligibility, reservation races and payout conservation.

• Demonstrate React/.NET integration, identity controls and failure recovery.

Engineering learning

• Compare Python and C# reference calculations and routing outputs.

• Rehearse no-money pilot procedures without building hardware.

• Make assumptions, failure denominators and unverified outcomes explicit.

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Useful testing without provider credentials.

An invented local HTTP contract is not a named provider sandbox.

Provider rehearsal

- Actual local HTTP timeout after commit and ambiguous 502 response tested.
- Reconcile by lookup instead of blindly repeating financial submission.
- Idempotency, callback duplication, wrong amount and reversal tests passed.

Calibration rehearsal

- Seven synthetic commodity pipelines require explicit synthetic opt-in.
- Repeated lot IDs across train/holdout are rejected.
- Candidate reports do not alter runtime Q10 or establish biological validity.

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From initial demo to reviewed software acceptance.

Work-package groupings, not invented calendar-week timesheets.

Foundation and corrections

• Initial FastAPI/React demo and five portals; cold-start transport fixes.

• Claims audit replaced unsupported recovery/field assertions.

• Manual confirmation, eligibility and bounded fallback became explicit.

Hardening and latest work

• v2: access, checkpoints, calibration/pilot tools and calculation parity.

• v3–v4: native workflow, auth, routing, recovery, offline lab and deployment checks.

• Current: fresh two-role browser journey, identity correction and safe consolidation.

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A tested prototype is not a certified deployment.

Hardware fabrication remains excluded, not a pending task.

Prepared / not executed here

• Docker/Compose and container-build/image-scan CI are supplied.

• No Docker runtime/daemon is available in this sandbox; image scan unexecuted.

• Cloud/TLS deployment, external security review and operations sign-off remain open.

Requires external evidence

• Select a permitted provider product and obtain test access for live API validation.

• Collect genuine controlled quality observations and independent holdout lots.

• Obtain operator/owner consent, quality review and actual outcome evidence before a trial.

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An inspectable project—not a claimed physical rescue.

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Delivered project artifacts

- Editable 17-slide PPTX and matching slide PDF.
- Seven-page project technical/research report; no peer-review claim.
- One master README, source, essential tests and retained evidence.

Read the results correctly

- Implemented: tested software paths and recorded state transitions.
- Simulated: telemetry inputs, balances, insurance and farmer credits.
- Unverified: real delivery benefit, biological accuracy and live-provider acceptance.